

ORIGINAL ARTICLE

Infrared thermography as a technique to measure physiological stress in birds: Body region and image angle matter

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Abstract

In vertebrates, changes in surface temperature following exposure to an acute stressor are thought to be promising indicators of the physiological stress response that may be captured noninvasively by infrared thermography. However, the efficacy of using stress-induced changes in surface temperature as indicators of physiological stress-responsiveness requires: (1) an understanding of how such responses vary across the body, (2) a magnitude of local, stress-induced thermal responses that is large enough to discriminate and quantify differences among individuals with conventional technologies, and (3) knowledge of how susceptible measurements across different body regions are to systematic error. In birds, temperature of the bare tissues surrounding the eye (the periorbital, or “eye,” region) and covering the bill have each been speculated as possible predictors of stress physiological state. Using the domestic pigeon (*Columba livia domestica*; $n = 9$), we show that stress-induced changes in surface temperature are most pronounced at the bill and that thermal responses at only the bill have sufficient resolution to detect and quantify differences in responsiveness among individuals. More importantly, we show that surface temperature estimates at the eye region experience greater error due to changes in bird orientation than those at the bill. Such error concealed detection of stress-induced thermal responses at the eye region. Our results highlight that: (1) in some species, bill temperature may serve as a more robust indicator of autonomic stress-responsiveness than eye region temperature, and (2) future studies should account for spatial orientation of study individuals if inference is to be drawn from infrared thermographic images.

KEYWORDS

autonomic nervous system, infrared thermography, stress, thermoregulation

1 | INTRODUCTION

An influence of acute stress exposure on body temperature in humans has been known to the medical community for

centuries (Yeo, 2005). Over the past 50 years, observations of stress-induced changes in body temperature across various vertebrates (reviewed elsewhere; Oka et al., 2001) have suggested that this phenomenon may be exploited to approximate

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